gpeyt1-01 · SPT (N1)60

GEO PE · Long Calculation Problems I — Topic 1: Site Characterization

Problem

A boring for a highway embankment encounters 8 ft of moist silty sand (γ = 112 pcf) over saturated clean fine sand (γsat = 124 pcf). The water table is at 8 ft. At a depth of 22 ft the field SPT blow count is N = 14 blows/ft. The automatic hammer was calibrated at an energy ratio of 80%. The driller also reports a 2 ft sample recovery, a 140 lb hammer with a 30 in. drop, and AWJ rods (none of which need further correction for this problem).

Most nearly, the corrected blow count (N1)60 at 22 ft is:

A. 20
B. 17
C. 15
D. 11

Given equations / data

> γw = 62.4 pcf; N60 = (Eeff/60)·Nmeas; (N1)60 = CN·N60; CN = 0.77 log10(20/Po), Po = vertical effective stress in tsf

Worked solution

1. Po = [8(112) + 14(124 − 62.4)]/2000 = 0.879 tsf
2. N60 = 14×80/60 = 18.67
3. CN = 0.77 log10(20/0.879) = 1.045
4. (N1)60 = 18.67×1.045 = 19.5 ≈ 20

Final answer

20

Why the other values are wrong

- **17** — used total stress for Po
- **15** — skipped the energy correction
- **11** — inverted the energy ratio (60/80)

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.8.1.1–3.8.1.2 (Hammer Efficiency, Depth Adjustment), p. 124

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=NwQ24-zlxyU, https://www.youtube.com/watch?v=DJaHlgyaOAg (no creator values copied).

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gpeyt1-02 · RQD

GEO PE · Long Calculation Problems I — Topic 1: Site Characterization

Problem

A 5.0 ft (60 in.) NQ core run in a jointed limestone recovers the following intact pieces (in., measured along the core axis): 5.5, 2.0, 7.0, 3.5, 4.0, 1.5, 9.0, 3.0, 6.5, 2.5, 4.5. The rest of the run was lost as rubble. The core barrel was double-tube, drilling took 18 minutes, and the drill water return was 80%.

Most nearly, the RQD of this run and its rock-quality description are:

A. 82%, good
B. 54%, fair
C. 74%, fair
D. 61%, fair

Given equations / data

> RQD = Σ(length of intact pieces ≥ 4 in.)/(total core run length) × 100%

Worked solution

1. Pieces ≥ 4 in.: 5.5, 7.0, 4.0, 9.0, 6.5, 4.5 → Σ = 36.5 in.
2. RQD = 36.5/60 × 100 = 60.8%
3. Recovery = 49.0/60 = 82% (not RQD)
4. RQD 61% → fair (handbook RQD description table)

Final answer

61%, fair

Why the other values are wrong

- **82%, good** — reported total core recovery (all pieces) as RQD
- **54%, fair** — excluded pieces exactly 4 in. long
- **74%, fair** — divided by recovered length instead of run length

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.7.4.1 Rock Quality Designation (RQD), p. 120

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=8cX88YWUo4U (no creator values copied).

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gpeyt1-03 · USCS from gradation

GEO PE · Long Calculation Problems I — Topic 1: Site Characterization

Problem

A sieve analysis on a split-spoon sample gives: 96% passing No. 4, 3% passing No. 200, D10 = 0.12 mm, D30 = 0.3 mm, D60 = 0.95 mm. The sample is non-plastic, brown, moist, with a natural water content of 9%. The boring log also notes trace mica.

Most nearly, Cu, Cc and the USCS group symbol are:

A. Cu = 7.9, Cc = 1.27, SW
B. Cu = 7.9, Cc = 0.79, SP-SM
C. Cu = 7.9, Cc = 0.79, SP
D. Cu = 7.9, Cc = 0.79, SW

Given equations / data

> Cu = D60/D10; Cc = D30²/(D10·D60); SW requires Cu ≥ 6 and 1 ≤ Cc ≤ 3; fines < 5%

Worked solution

1. Cu = 0.95/0.12 = 7.92 (≥ 6 OK)
2. Cc = 0.3²/(0.12×0.95) = 0.789 (< 1, fails)
3. 96% passes No. 4 → sand; 3% fines < 5% → clean sand
4. Fails Cc → poorly graded sand, SP

Final answer

Cu = 7.9, Cc = 0.79, SP

Why the other values are wrong

- **Cu = 7.9, Cc = 1.27, SW** — inverted Cc
- **Cu = 7.9, Cc = 0.79, SP-SM** — treated 3% fines as a dual-symbol (5–12%) soil
- **Cu = 7.9, Cc = 0.79, SW** — checked Cu only

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.7.2 USCS, p. 110–117; §3.8.4 Gradation Tests, p. 136

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gpeyt1-04 · Atterberg / LI

GEO PE · Long Calculation Problems I — Topic 1: Site Characterization

Problem

A Shelby-tube sample of gray clay from 30 ft has LL = 58, PL = 24, and natural water content w = 52%. The torvane reading was 0.35 tsf, the pocket penetrometer 0.5 tsf, and 100% passes the No. 200 sieve. The engineer wants the liquidity index and a USCS fine-grained symbol from the plasticity chart.

Most nearly, the LI and USCS symbol are:

A. LI = 0.18, CH
B. LI = 0.82, MH
C. LI = 0.48, CL
D. LI = 0.82, CH

Given equations / data

> PI = LL − PL; LI = (w − PL)/PI; A-line: PI = 0.73(LL − 20)

Worked solution

1. PI = 58 − 24 = 34
2. LI = (52 − 24)/34 = 0.824
3. A-line at LL = 58: 0.73(58−20) = 27.7 < PI = 34 → above A-line
4. LL ≥ 50 and above A-line → CH; LI near 0.9 → soft, near-liquid consistency

Final answer

LI = 0.82, CH

Why the other values are wrong

- **LI = 0.18, CH** — computed consistency index (LL − w)/PI
- **LI = 0.82, MH** — read the soil as below the A-line
- **LI = 0.48, CL** — divided by LL and used LL < 50 boundary

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.8.2 Atterberg Limits, p. 132; §3.7.2 plasticity chart

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gpeyt1-05 · (N1)60 profile

GEO PE · Long Calculation Problems I — Topic 1: Site Characterization

Problem

A sand deposit has γ = 118 pcf above the water table at 6 ft and γsat = 128 pcf below. Two SPT samples were taken with a safety hammer (ER = 70%): N = 11 at 12 ft and N = 24 at 45 ft. The geologist claims the deeper sand is 'much denser' because its raw N is ~65% higher. The rig used 2 in. OD split spoon and 1.4 in. ID liner-free sampler.

Most nearly, (N1)60 at 45 ft is:

A. 28
B. 18
C. 23
D. 15

Given equations / data

> N60 = (ER/60)·N; CN = 0.77 log10(20/Po), Po in tsf effective; γw = 62.4 pcf

Worked solution

1. Po(12) = 0.551 tsf → (N1)60 = 11×70/60×1.201 = 15.4
2. Po(45) = [6(118) + 39(128−62.4)]/2000 = 1.633 tsf
3. CN = 0.77 log10(20/1.633) = 0.838
4. (N1)60 = 24×70/60×CN = 23.5; after correction the two sands are much closer than raw N suggests

Final answer

23

Why the other values are wrong

- **28** — forgot the overburden correction
- **18** — used total stress in CN
- **15** — reported the 12 ft value

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.8.1.1–3.8.1.2, p. 124

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gpeyt1-06 · OC clay settlement

GEO PE · Long Calculation Problems I — Topic 2: Soil Mechanics, Lab Testing & Analysis

Problem

A warehouse slab will add a uniform Δσ = 1,200 psf (wide load) at the top of a 16 ft clay layer. Profile: 0–10 ft sand, γ = 115 pcf (water table at 10 ft, top of clay); 10–26 ft clay, γsat = 108 pcf, e0 = 1.05, Cc = 0.42, Cr = 0.06, σ′p = 1,450 psf; below is dense gravel. The clay's cv = 0.35 ft²/day and LL = 61 (not needed here). Use one layer evaluated at mid-depth.

Most nearly, the primary consolidation settlement of the clay is:

A. 10.6 in.
B. 10.0 in.
C. 1.4 in.
D. 8.0 in.

Given equations / data

> γw = 62.4 pcf; OC clay with σ′0 < σ′p < σ′f: S = [Cr H/(1+e0)] log(σ′p/σ′0) + [Cc H/(1+e0)] log(σ′f/σ′p)

Worked solution

1. σ′0 at mid-clay = 10(115) + 8(108 − 62.4) = 1,515 psf
2. σ′f = 1,515 + 1,200 = 2,715 psf > σ′p = 1,450 → crosses σ′p
3. Recompression: 0.06×16/2.05×log(1450/1515) = -0.0089 ft
4. Virgin: 0.42×16/2.05×log(2715/1450) = 0.8928 ft
5. S = 0.8840 ft = 10.6 in.

Final answer

10.6 in.

Why the other values are wrong

- **10.0 in.** — treated the clay as normally consolidated (Cc from σ′0)
- **1.4 in.** — used Cr for the whole stress range
- **8.0 in.** — used total stress at mid-clay

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.2.1–3.2.2 Consolidation (NC and OC soils), pp. 82–84

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=IGS94AWmPWA, https://www.youtube.com/watch?v=NwGztxby-vQ (no creator values copied).

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gpeyt1-07 · Time rate of consolidation

GEO PE · Long Calculation Problems I — Topic 2: Soil Mechanics, Lab Testing & Analysis

Problem

An 18 ft layer of soft clay lies between a sand blanket above and a fractured, free-draining sandstone below. A preload is placed instantly. Lab data: cv = 0.28 ft²/day, Cc = 0.38, e0 = 1.12. Ultimate primary settlement under the preload is estimated at 9.6 in. The contractor wants to know how long the preload must stay to reach 90% average consolidation.

Most nearly, the time for U = 90% is:

A. 981 days
B. 57 days
C. 27 days
D. 245 days

Given equations / data

> t = Tv Hd²/cv; Hd = longest drainage distance; Tv(U = 90%) = 0.848, Tv(50%) = 0.197

Worked solution

1. Double drainage (sand top, free-draining rock base) → Hd = H/2 = 9 ft
2. t90 = 0.848 × 9² / 0.28 = 245 days (≈ 0.67 yr)
3. Settlement at that time ≈ 0.9 × 9.6 = 8.6 in. (extra data not needed)

Final answer

245 days

Why the other values are wrong

- **981 days** — used Hd = full thickness (single drainage)
- **57 days** — used Tv for 50%
- **27 days** — forgot to square Hd

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.2.3 Time Rate of Settlement and U–Tv table, p. 85

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gpeyt1-08 · CD triaxial strength

GEO PE · Long Calculation Problems I — Topic 2: Soil Mechanics, Lab Testing & Analysis

Problem

A consolidated-drained (CD) triaxial test on a remolded clean sand ($c' = 0$) consolidated to $\sigma'_3 = 20$ psi fails at a deviator stress of 52 psi. Pore pressure was zero throughout; specimen diameter 2.8 in., height 5.6 in.; axial strain at failure 6%. A second specimen of the same sand will be tested at $\sigma'_3 = 35$ psi.

Most nearly, the deviator stress at failure expected for the 35 psi test is:

A. 91 psi
B. 126 psi
C. 99 psi
D. 45 psi

Given equations / data

> $\sin\varphi' = (\sigma'_1 - \sigma'_3)/(\sigma'_1 + \sigma'_3)$; $\sigma'_1 = \sigma'_3 \tan^2(45 + \varphi'/2)$ for $c' = 0$

Worked solution

1. Test 1: $\sigma'_1 = 20 + 52 = 72$ psi
2. $\sin\varphi' = 52/92 = 0.5652 \rightarrow \varphi' = 34.4°$
3. Kp-type ratio $\tan^2(45 + \varphi'/2) = 3.600$
4. Test 2: $\sigma'_1 = 35 \times 3.600 = 126.0$ psi; $\Delta\sigma_f = 126.0 - 35 = 91.0$ psi

Final answer

91 psi

Why the other values are wrong

- **126 psi** — reported σ'_1 instead of deviator stress
- **99 psi** — used $\tan\varphi = \Delta\sigma/\sigma_1$
- **45 psi** — reported the Mohr-circle radius (½ deviator)

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.3.2 Shear Strength — Effective Stress (Mohr–Coulomb), p. 87

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=ZUGdBao3tkc (no creator values copied).

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gpeyt1-09 · Falling-head k

GEO PE · Long Calculation Problems I — Topic 2: Soil Mechanics, Lab Testing & Analysis

Problem

A falling-head permeability test is run on a silty sand specimen 2.5 in. in diameter and 6.0 in. long. The standpipe inside diameter is 0.25 in. The head drops from 40 in. to 24 in. in 26 min. Water temperature 20°C; specimen dry unit weight 104 pcf; Gs = 2.66.

Most nearly, the hydraulic conductivity k is:

A. 5.0e-05 cm/s
B. 2.2e-05 cm/s
C. 2.0e-05 cm/s
D. 5.0e-04 cm/s

Given equations / data

> k = (a L)/(A t) · ln(h1/h2); a = standpipe area, A = specimen area

Worked solution

1. a = π/4 (0.25)² = 0.04909 in.²; A = π/4 (2.5)² = 4.9087 in.²
2. t = 26 × 60 = 1560 s
3. k = (0.04909×6.0)/(4.9087×1560) × ln(40/24) = 1.965e-05 in./s
4. × 2.54 = 4.99e-05 cm/s

Final answer

5.0e-05 cm/s

Why the other values are wrong

- **2.2e-05 cm/s** — used log10 instead of ln
- **2.0e-05 cm/s** — left the result in in./s but labeled cm/s
- **5.0e-04 cm/s** — used diameter ratio d/D instead of area ratio a/A

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.8.5 Permeability Testing, p. 139

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=RqaahrVtV4k (no creator values copied).

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gpeyt1-10 · Δσ′ from drawdown

GEO PE · Long Calculation Problems I — Topic 2: Soil Mechanics, Lab Testing & Analysis

Problem

A uniform silty sand deposit is 50 ft thick over rock. Above the water table γ = 110 pcf (moist); below, γsat = 122 pcf. Dewatering for a nearby excavation will lower the water table from 4 ft to 14 ft below grade over several months. Assume the sand above the new water table drains to the moist unit weight and no capillary rise. A settlement-sensitive pipeline rests on piles tipped at 40 ft.

Most nearly, the increase in vertical effective stress at 40 ft is:

A. 624 psf
B. 504 psf
C. 120 psf
D. 1,100 psf

Given equations / data

> σ′ = σ − u; u = γw(z − zw); γw = 62.4 pcf

Worked solution

1. Before: σ′ = 4(110) + 36(122−62.4) = 2,586 psf
2. After: σ′ = 14(110) + 26(122−62.4) = 3,090 psf
3. Δσ′ = 504 psf = 10 ft × (110 − 122 + 62.4)

Final answer

504 psf

Why the other values are wrong

- **624 psf** — took Δσ′ = γw Δzw (ignored the drop in total stress)
- **120 psf** — used only (γsat − γmoist)Δz
- **1,100 psf** — added only the moist weight and forgot buoyancy before

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.3 Effective and Total Stresses, p. 87

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=ZUGdBao3tkc (no creator values copied).

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gpeyt1-11 · OSHA sloping

GEO PE · Long Calculation Problems I — Topic 3: Construction Observation, QA/QC & Safety

Problem

A 13 ft deep utility trench with a 5 ft wide bottom will be sloped (no shoring) on both sides through a medium-stiff silty clay classified by the competent person as OSHA Type B (qu = 1.0 tsf, fissured, previously disturbed). The trench run is 240 ft long. The spoil pile will be kept 2 ft back from the edge, and the excavator weighs 48,000 lb.

Most nearly, the minimum top width of the trench is:

A. 24 ft
B. 44 ft
C. 31 ft
D. 18 ft

Given equations / data

> OSHA maximum allowable slopes (H:V): Type A 3/4:1, Type B 1:1, Type C 1½:1 (excavations ≤ 20 ft)

Worked solution

1. Type B → 1H:1V on each side
2. Horizontal run each side = 13 × 1 = 13 ft
3. Top width = 5 + 2(13) = 31.0 ft
4. (Excavated volume ≈ (5+31.0)/2 × 13 × 240/27 = 2,080 yd³ — not asked)

Final answer

31 ft

Why the other values are wrong

- **24 ft** — used the Type A slope 3/4:1
- **44 ft** — used the Type C slope 1½:1
- **18 ft** — sloped only one side

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.10.2 Slope and Shield Configurations (Allowable Slopes table), p. 152; 29 CFR 1926 Subpart P App. B

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=hE7KZATybxA, https://www.youtube.com/watch?v=5IweGa4yOBc (no creator values copied).

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gpeyt1-12 · Sand-cone relative compaction

GEO PE · Long Calculation Problems I — Topic 3: Construction Observation, QA/QC & Safety

Problem

A field density test on a compacted embankment lift: the mass of calibrated sand that filled the hole (after subtracting the cone correction) was 3.36 lb; the bulk density of the calibrated sand is 93.5 pcf. The moist soil removed from the hole weighed 4.66 lb, and its oven-dry water content is 11.8%. The lab standard Proctor gave $\gamma_{d,max}$ = 121.0 pcf at w_{opt} = 10.5%. Specification: ≥ 95% relative compaction, w within −1/+3% of optimum. Lift thickness 8 in.

Most nearly, the relative compaction and the test result are:

A. 107.2%, passes
B. 94.5%, fails
C. 95.9%, passes
D. 104.3%, passes

Given equations / data

> $V = W_{sand}/\gamma_{sand}$; $\gamma_t = W/V$; $\gamma_d = \gamma_t/(1 + w)$; $RC = \gamma_d/\gamma_{d,max} \times 100\%$

Worked solution

1. Hole volume $V = 3.36/93.5 = 0.0359$ ft³
2. $\gamma_t = 4.66/0.0359 = 129.7$ pcf
3. $\gamma_d = 129.7/1.118 = 116.0$ pcf
4. $RC = 116.0/121.0 = 95.9\%$ (≥ 95% required); w = 11.8% is +1.3% of opt (within −1/+3) → passes

Final answer

95.9%, passes

Why the other values are wrong

- **107.2%, passes** — used moist unit weight (forgot ÷(1+w))
- **94.5%, fails** — multiplied by (1 − w) instead of dividing by (1 + w)
- **104.3%, passes** — inverted the ratio

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.9.1–3.9.2 Laboratory and Field Compaction, pp. 146–148; §3.8.3 Weight-Volume Relationships

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=v9reh5FJnUc, https://www.youtube.com/watch?v=v_1tQfxBowQ (no creator values copied).

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gpeyt1-13 · Compaction production

GEO PE · Long Calculation Problems I — Topic 3: Construction Observation, QA/QC & Safety

Problem

A contractor will compact 18,000 compacted yd³ of embankment with a single smooth-drum vibratory roller. Drum compacted width per pass = 7.0 ft; average speed = 2.5 mph; compacted lift = 8 in.; job efficiency = 80% (48 min/hr); test strip shows 6 passes are needed for 95% RC. Shrinkage from bank to compacted is 15% and the haul fleet delivers 300 bank yd³/hr.

Most nearly, how many roller-hours are required?

A. 10 hr
B. 47 hr
C. 59 hr
D. 50 hr

Given equations / data

> Compacted yd³/hr = 16.3 · W · S · L · efficiency / n (W ft, S mph, L in.)

Worked solution

1. Rate = 16.3 × 7.0 × 2.5 × 8 × 0.8 / 6 = 304.3 compacted yd³/hr
2. Hours = 18,000/304.3 = 59 hr
3. Shrinkage only converts compacted to bank yd³ — not needed because quantity is already compacted

Final answer

59 hr

Why the other values are wrong

- **10 hr** — multiplied by n instead of dividing
- **47 hr** — omitted the efficiency factor
- **50 hr** — applied the shrinkage factor to compacted quantity

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §2.3.3.1 Production Rate for Soil Compaction (Ch. 2 Construction)

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=v9reh5FJnUc (no creator values copied).

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gpeyt1-14 · Moisture conditioning

GEO PE · Long Calculation Problems I — Topic 3: Construction Observation, QA/QC & Safety

Problem

Borrow for a 52,000 ft³ (compacted) highway embankment has an in-situ water content of 7.5%. The specification requires placement at 13.0% and a dry unit weight of 114 pcf. The water truck holds 4,000 gal. The borrow's in-place moist unit weight is 118 pcf and Gs = 2.68 (not needed).

Most nearly, the volume of water that must be added is:

A. 39,140 gal
B. 92,514 gal
C. 5,225 gal
D. 40,514 gal

Given equations / data

> Gallons = γd,desired × Vcompacted × (wdesired − wborrow)/100 / 8.33 lb/gal

Worked solution

1. Δw = 13.0 − 7.5 = 5.5%
2. Water weight = 114 × 52,000 × 0.055 = 326,040 lb
3. Gallons = /8.33 = 39,140 gal (≈ 10 truckloads)

Final answer

39,140 gal

Why the other values are wrong

- **92,514 gal** — used the full desired w (ignored borrow moisture)
- **5,225 gal** — divided by 62.4 (got ft³, labeled gal)
- **40,514 gal** — used the moist borrow unit weight

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §2.3.1 Earthwork/compaction — amount of water to add (Ch. 2 Construction)

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=v_1tQfxBowQ, https://www.youtube.com/watch?v=f02U9lj1I58 (no creator values copied).

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gpeyt1-15 · Piezometer QA

GEO PE · Long Calculation Problems I — Topic 3: Construction Observation, QA/QC & Safety

Problem

A vibrating-wire piezometer is installed at the middle of a soft clay layer (tip at El. 60.0) beneath a staged embankment. Immediately after stage-1 fill, the excess pore pressure at the tip was 1,150 psf. Three months later the piezometer indicates a total head equivalent to a water level at El. 96.5, while the static groundwater table is at El. 88.0. Settlement plates show 7.2 in. of settlement; inclinometers show <0.3 in. lateral movement.

Most nearly, the local degree of consolidation at the piezometer tip is:

A. 54%
B. 46%
C. 76%
D. 19%

Given equations / data

> ue = γw × (piezometric elevation − static water elevation); Uz = 1 − ue/ue0; γw = 62.4 pcf

Worked solution

1. Excess head = 96.5 − 88.0 = 8.5 ft
2. ue = 8.5 × 62.4 = 530 psf
3. Uz = 1 − 530/1,150 = 0.539 ≈ 54%

Final answer

54%

Why the other values are wrong

- **46%** — reported ue/ue0 (fraction still remaining)
- **76%** — measured excess head from the ground surface (El. 92.0) instead of the static water table
- **19%** — used the clay unit weight (110 pcf) instead of γw

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.11 Geotechnical Instrumentation, p. 156; §3.2.3 consolidation

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=SXA_TC3XEwk (no creator values copied).

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gpeyt1-16 · Cyclic stress ratio

GEO PE · Long Calculation Problems I — Topic 4: Earthquake Engineering & Dynamic Loads

Problem

A bridge abutment site is underlain by loose to medium dense clean sand. Water table at 6 ft. γ = 112 pcf above and γsat = 125 pcf below. The design earthquake has Mw = 7.0 and a site-adjusted peak ground acceleration amax = 0.32g (PGA on rock was 0.24g). Evaluate the sand at 24 ft depth using the Seed–Idriss simplified procedure with the mean rd relation.

Most nearly, the CSR at 24 ft is:

A. 0.338
B. 0.196
C. 0.319
D. 0.239

Given equations / data

> CSR = 0.65 (amax/g)(σvo/σ′vo) rd; rd = 1.0 − 0.00233z for z < 30 ft (z in ft); γw = 62.4 pcf

Worked solution

1. σvo = 6(112) + 18(125) = 2,922 psf
2. u = 18(62.4) = 1,123 psf → σ′vo = 1,799 psf
3. rd = 1 − 0.00233(24) = 0.9441
4. CSR = 0.65 × 0.32 × (2,922/1,799) × 0.9441 = 0.319

Final answer

0.319

Why the other values are wrong

- **0.338** — omitted rd
- **0.196** — omitted σvo/σ′vo
- **0.239** — used rock PGA instead of site amax

References (Jayce's library)

- FHWA-NHI-11-032 (GEC 3) Eq. 6-16, §6.4 Simplified Procedure; rd relation (Liao & Whitman) in GEC 3

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=Hg3MTAvoJWw, https://www.youtube.com/watch?v=f9PIvbPStrU (no creator values copied).

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gpeyt1-17 · FS against liquefaction

GEO PE · Long Calculation Problems I — Topic 4: Earthquake Engineering & Dynamic Loads

Problem

A silty sand layer (FC = 12%) at 18 ft has a clean-sand-equivalent (N1)60cs = 13, giving CRR(M7.5) = 0.17 from the base curve. The design event is Mw = 6.75 with amax = 0.28g; the magnitude scaling factor for that event is MSF = 1.19. Kσ = 1.0. Water table 8 ft; γ = 115 pcf above, γsat = 122 pcf below. Use rd = 1 − 0.00233z.

Most nearly, the factor of safety against liquefaction is:

A. 0.69
B. 1.22
C. 1.16
D. 0.82

Given equations / data

> CSR = 0.65 (amax/g)(σvo/σ′vo) rd; FSL = (CRR7.5 · MSF · Kσ)/CSR

Worked solution

1. σvo = 8(115) + 10(122) = 2,140 psf; σ′vo = 1,516 psf
2. rd = 0.9581; CSR = 0.65×0.28×1.412×0.9581 = 0.246
3. FSL = 0.17×1.19×1.0/0.246 = 0.82 (<1 → liquefiable)

Final answer

0.82

Why the other values are wrong

- **0.69** — omitted MSF
- **1.22** — inverted FS (CSR/CRR)
- **1.16** — used total stress for σ′vo (ratio = 1)

References (Jayce's library)

- FHWA-NHI-11-032 (GEC 3) Ch. 6 Eq. 6-16 and FS definition, MSF discussion

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=Hg3MTAvoJWw (no creator values copied).

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gpeyt1-18 · Mononobe–Okabe

GEO PE · Long Calculation Problems I — Topic 4: Earthquake Engineering & Dynamic Loads

Problem

An 18 ft cantilever wall retains level, dry granular backfill (γ = 120 pcf, φ = 34°, c = 0). The back face is vertical and wall friction is neglected (δ = 0). Seismic design uses kh = 0.16, kv = 0. The wall has a 10 ft base and a 2 ft thick stem; passive resistance at the toe is ignored.

Most nearly, the total (static + seismic) active thrust PAE is:

A. 7,403 lb/ft
B. 5,496 lb/ft
C. 14,807 lb/ft
D. 6,219 lb/ft

Given equations / data

> Mononobe–Okabe with δ = 0, β = 0, i = 0: $\theta = \tan^{-1}[kh/(1-kv)]$; $KAE = \cos^2(\varphi-\theta)/\{\cos^2\theta\,[1+\sqrt{(\sin\varphi\,\sin(\varphi-\theta)/\cos\theta)}]^2\}$; $PAE = \tfrac{1}{2}\,KAE\,\gamma\,H^2\,(1-kv)$

Worked solution

1. $\theta = \tan^{-1}(0.16) = 9.09°$
2. $KAE = \cos^2(34-9.09)/\{\cos^2(9.09)[1+\sqrt{(\sin34\cdot\sin24.91/\cos9.09)}]^2\} = 0.381$
3. (static Ka = 0.283)
4. $PAE = \tfrac{1}{2}(0.381)(120)(18)^2 = 7{,}403$ lb/ft

Final answer

7,403 lb/ft

Why the other values are wrong

- **5,496 lb/ft** — used static Rankine Ka
- **14,807 lb/ft** — forgot the ½
- **6,219 lb/ft** — multiplied by (1 − kh) instead of (1 − kv)

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.1.5 Pseudostatic Analysis and Earthquake Loads (Mononobe–Okabe), p. 79; GEC 3 §11.3.1

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=w48hBSXrUhY (no creator values copied).

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gpeyt1-19 · Pseudostatic rock slope

GEO PE · Long Calculation Problems I — Topic 4: Earthquake Engineering & Dynamic Loads

Problem

A highway rock cut has a daylighting planar joint dipping α = 35° with length L = 60 ft along the joint per ft of cut. The sliding block weighs W = 180,000 lb/ft. Joint strength: c = 500 psf, φ = 30°. The slope is fully drained (no water in the tension crack or on the joint). For seismic screening use k = 0.15. The rock unit weight is 160 pcf and RQD is 72% (not needed).

Most nearly, the pseudostatic factor of safety is:

A. 1.12
B. 0.85
C. 0.92
D. 1.18

Given equations / data

> FS = [cL + (W cos α − kW sin α − PWR − PWJ sin α) tan φ]/(W sin α + kW cos α + PWJ cos α); PWR = PWJ = 0 when drained

Worked solution

1. cL = 500×60 = 30,000 lb/ft
2. Normal: W cos α − kW sin α = 147,447 − 15,487 = 131,961 lb/ft
3. Resisting = 30,000 + 131,961×tan30 = 106,188
4. Driving = W sin α + kW cos α = 125,361
5. FS = 0.85 (static 1.12)

Final answer

0.85

Why the other values are wrong

- **1.12** — static FS (dropped k)
- **0.92** — added k only to the driving side
- **1.18** — inverted FS

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.6.3 Rock Slope Failure (Stability of Rock Slope figure), p. 104

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=90zXmI_yn_g (no creator values copied).

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gpeyt1-20 · Pseudostatic infinite slope

GEO PE · Long Calculation Problems I — Topic 4: Earthquake Engineering & Dynamic Loads

Problem

A long, dry, cohesionless fill slope is inclined at β = 24° (about 2.25H:1V) with φ′ = 36°, γ = 120 pcf. A shallow failure surface parallel to the slope at 6 ft depth is being checked. The seismic coefficient is kh = 0.12 applied horizontally (out of slope) at the block's center of mass. No pore pressure.

Most nearly, the pseudostatic factor of safety is:

A. 1.63
B. 1.46
C. 1.35
D. 1.22

Given equations / data

> Static infinite slope (ru = 0): F = tan φ′/tan β. Resolve W and khW parallel/normal to the slip plane: F = (cos β − kh sin β) tan φ′/(sin β + kh cos β)

Worked solution

1. Static F = tan36/tan24 = 1.63
2. Normal (per W): cos24 − 0.12 sin24 = 0.8647
3. Driving (per W): sin24 + 0.12 cos24 = 0.5164
4. F = 0.8647×tan36/0.5164 = 1.22 (depth and γ cancel)

Final answer

1.22

Why the other values are wrong

- **1.63** — static FS only
- **1.46** — divided static FS by (1 + kh)
- **1.35** — added kh·sin β to the normal force (wrong sign)

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.6.4 Infinite Slope, p. 105; pseudostatic resolution as in §3.6.3 (kW terms); GEC 3 Ch. 7

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=SXA_TC3XEwk (no creator values copied).

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gpeyt1-21 · Borrow quantity

GEO PE · Long Calculation Problems I — Topic 5: Earth Structures, Ground Improvement & Pavements

Problem

A highway embankment needs 85,000 yd³ of compacted fill at 95% of standard Proctor γd,max = 118.5 pcf. The borrow pit soil has an in-place moist unit weight of 112 pcf at w = 14%. The loose (truck) unit weight is 96 pcf and each truck carries 14 loose yd³. Ignore haul losses.

Most nearly, the volume of borrow (bank measure) required is:

A. 85,436 yd³
B. 102,524 yd³
C. 74,181 yd³
D. 97,397 yd³

Given equations / data

> Dry solids are conserved: Vborrow = Vfill × γd,fill / γd,borrow; γd = γ/(1 + w)

Worked solution

1. γd,fill = 0.95 × 118.5 = 112.6 pcf
2. γd,borrow = 112/1.14 = 98.2 pcf
3. Vborrow = 85,000 × 112.6/98.2 = 97,397 yd³

Final answer

97,397 yd³

Why the other values are wrong

- **85,436 yd³** — used moist borrow unit weight as dry
- **102,524 yd³** — forgot the 95% factor
- **74,181 yd³** — inverted the density ratio

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.8.3 Weight-Volume Relationships, p. 133; §2.3.1 earthwork unit weights (γB, γC)

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=v9reh5FJnUc (no creator values copied).

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gpeyt1-22 · Infinite slope with pore pressure

GEO PE · Long Calculation Problems I — Topic 5: Earth Structures, Ground Improvement & Pavements

Problem

A long natural hillside of residual silty sand slopes at β = 18°. A shallow slip surface parallel to the slope is suspected at z = 8 ft (vertical depth). γ = 122 pcf, c′ = 0, φ′ = 30°. During wet season a piezometer on the slip surface reads u = 280 psf. The road at the toe carries 2,000 vehicles/day.

Most nearly, the factor of safety during the wet season is:

A. 1.78
B. 1.27
C. 1.21
D. 0.68

Given equations / data

> $r_u = u/(\gamma z)$; $F = [1 - r_u(1 + \tan^2\beta)] \tan\phi'/\tan\beta$

Worked solution

1. $r_u = 280/(122 \times 8) = 0.287$
2. $1 + \tan^2 18 = 1.1056$
3. $F = [1 - 0.287 \times 1.1056] \times \tan 30/\tan 18 = 0.6828 \times 1.777 = 1.21$

Final answer

1.21

Why the other values are wrong

- **1.78** — dry case (ignored u)
- **1.27** — omitted the (1 + tan²β) term
- **0.68** — used γw instead of γ in ru

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.6.4 Infinite Slope, p. 105

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gpeyt1-23 · Flexible pavement SN

GEO PE · Long Calculation Problems I — Topic 5: Earth Structures, Ground Improvement & Pavements

Problem

An AASHTO 1993 flexible pavement design gives a required structural number SN = 4.6 for 8.2 million ESALs (R = 95%, S0 = 0.45, ΔPSI = 1.7, subgrade MR = 6,000 psi). Layers: 4.0 in. HMA (a1 = 0.44); crushed-stone base (a2 = 0.14, m2 = 1.0); 8.0 in. granular subbase (a3 = 0.11, m3 = 0.9).

Most nearly, the minimum base thickness D2 is:

A. 14.0 in.
B. 14.6 in.
C. 20.3 in.
D. 16.3 in.

Given equations / data

```
SN = a1D1 + a2D2m2 + a3D3m3
```

Worked solution

1. a1D1 = 0.44×4.0 = 1.76
2. a3D3m3 = 0.11×8.0×0.9 = 0.792
3. D2 = (4.6 − 1.76 − 0.792)/(0.14×1.0) = 14.6 in. (round up in practice)

Final answer

14.6 in.

Why the other values are wrong

- **14.0 in.** — ignored the subbase drainage coefficient m3
- **20.3 in.** — forgot the subbase contribution
- **16.3 in.** — applied m3 = 0.9 to the base instead

References (Jayce's library)

- FHWA NHI-05-037 Geotechnical Aspects of Pavements, Eq. 5.40 (SN = a1D1 + a2D2m2 + a3D3m3); NCEES PE Civil Reference Handbook v2.1 §3.19 Pavements

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=uGGy0dkZaX4, https://www.youtube.com/watch?v=wBgMPZkg_IE (no creator values copied).

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gpeyt1-24 · Geotextile wall spacing

GEO PE · Long Calculation Problems I — Topic 5: Earth Structures, Ground Improvement & Pavements

Problem

A 16 ft geotextile-wrapped reinforced wall retains granular backfill (γ = 120 pcf, φ = 32°). A uniform surcharge q = 250 psf acts on top (no live point loads). The geotextile has wide-width Tult = 2,400 lb/ft and cumulative reduction factors ΠRF = 2.6. Use FS = 1.5 on reinforcement. Check the bottom layer at Z = 16 ft. Facing batter is vertical; foundation is dense gravel.

Most nearly, the maximum vertical spacing of the bottom layer is:

A. 2.40 ft
B. 1.04 ft
C. 0.09 ft
D. 0.92 ft

Given equations / data

```
σh = Ka γ Z + Ka q; Tallow = Tult/ΠRF; Sv = Tallow/(σh · FS); Ka = tan²(45 − φ/2)
```

Worked solution

1. Ka = tan²(45 − 32/2) = 0.307
2. σh = 0.307×120×16 + 0.307×250 = 667 psf
3. Tallow = 2,400/2.6 = 923 lb/ft
4. Sv = 923/(667×1.5) = 0.92 ft (≈ 11 in.)

Final answer

0.92 ft

Why the other values are wrong

- **2.40 ft** — used Tult without reduction factors
- **1.04 ft** — omitted the surcharge pressure
- **0.09 ft** — used Kp in place of Ka

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.13.5.1 Geotextile-Reinforced Walls, p. 170

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=4cCtjNpcFkA (no creator values copied).

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gpeyt1-25 · Geotextile length

GEO PE · Long Calculation Problems I — Topic 5: Earth Structures, Ground Improvement & Pavements

Problem

Same 16 ft geotextile wall: $\gamma = 120$ pcf, $\varphi = 32°$, $q = 250$ psf, FS = 1.5. Design a layer at $Z = 4$ ft below the top with $S_v = 2.0$ ft. Interface friction $\delta = \tfrac{2}{3}\varphi$, $c_a = 0$. Wall length $L = L_e + L_R$ (L_R measured from the face to the Rankine plane).

Most nearly, the total geotextile length L at Z = 4 ft is:

A. 1.8 ft
B. 10.2 ft
C. 4.0 ft
D. 8.4 ft

Given equations / data

$$\sigma_h = K_a(\gamma Z + q);\quad L_e = \frac{S_v\,\sigma_h\,FS}{2(c_a + \gamma Z \tan\delta)};\quad L_R = (H - Z)\tan\left(45 - \frac{\varphi}{2}\right)$$

Worked solution

1. $\sigma_h = 0.307(120\times4 + 250) = 224$ psf
2. $\delta = 21.33°$; $L_e = \dfrac{2.0\times224\times1.5}{2\times120\times4\times\tan21.33} = 1.79$ ft
3. $L_R = (16 - 4)\tan(45 - 16) = 6.65$ ft
4. $L = 1.79 + 6.65 = 8.4$ ft (a practical minimum length would also be checked)

Final answer

8.4 ft

Why the other values are wrong

- **1.8 ft** — reported L_e only (forgot L_R)
- **10.2 ft** — used one face of friction (dropped the 2)
- **4.0 ft** — measured L_R with Z instead of $H - Z$

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.13.5.1 Geotextile-Reinforced Walls, p. 170

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=4cCtjNpcFkA (no creator values copied).

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gpeyt1-26 · Flow-net seepage

GEO PE · Long Calculation Problems I — Topic 6: Groundwater & Seepage

Problem

A concrete diversion weir on a silty sand foundation (k = 2.5e-04 cm/s, isotropic) has headwater 34 ft and tailwater 6 ft above the foundation surface. A properly drawn flow net beneath the weir has Nf = 4 flow channels and Nd = 14 equipotential drops. The weir is 450 ft long. The sheet-pile cutoff is 20 ft deep and the sand is 45 ft thick.

Most nearly, the total seepage under the weir is:

A. 3,098 ft³/day
B. 31,252 ft³/day
C. 2,551 ft³/day
D. 6 ft³/day

Given equations / data

> q = k H Nf/Nd per unit length; H = total head loss; 1 cm/s = 2,835 ft/day

Worked solution

1. k = 2.5e-04 cm/s × 2,835 = 0.709 ft/day
2. H = 34 − 6 = 28 ft
3. q = 0.709 × 28 × 4/14 = 5.67 ft³/day/ft
4. Q = 5.67 × 450 = 2,551 ft³/day

Final answer

2,551 ft³/day

Why the other values are wrong

- **3,098 ft³/day** — used headwater depth instead of head difference
- **31,252 ft³/day** — inverted Nf/Nd
- **6 ft³/day** — reported per-foot flow (forgot length)

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.16.4 Flow Nets, p. 190; §3.16.1 Darcy's Law

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=RqaahrVtV4k, https://www.youtube.com/watch?v=ng3In5ayyaE, https://www.youtube.com/watch?v=_qeKY4icSPs (no creator values copied).

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gpeyt1-27 · Exit gradient / boiling

GEO PE · Long Calculation Problems I — Topic 6: Groundwater & Seepage

Problem

Inside a sheet-pile cofferdam excavated in fine sand (γsat = 124 pcf, Gs = 2.66, e = 0.62), a flow net shows the last equipotential square adjacent to the excavation floor has a head loss of Δh = 2.8 ft over a flow-path length of 4.5 ft. Total head difference across the cofferdam is 16 ft and the sheet piles are embedded 14 ft below the floor.

Most nearly, the factor of safety against boiling at the exit is:

A. 3.19
B. 1.59
C. 0.63
D. 0.86

Given equations / data

> icr = γ′/γw; iexit = Δh/l; FS = icr/iexit; γw = 62.4 pcf

Worked solution

1. γ′ = 124 − 62.4 = 61.6 pcf → icr = 0.987
2. (check: (Gs−1)/(1+e) = 1.025)
3. iexit = 2.8/4.5 = 0.622
4. FS = 0.987/0.622 = 1.59

Final answer

1.59

Why the other values are wrong

- **3.19** — used γsat/γw for icr
- **0.63** — inverted FS
- **0.86** — used total head/embedment as the exit gradient

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.16.3 Flow Through Soil / §3.16.4 Flow Nets, pp. 188–190

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=RqaahrVtV4k (no creator values copied).

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gpeyt1-28 · Unconfined well (Thiem)

GEO PE · Long Calculation Problems I — Topic 6: Groundwater & Seepage

Problem

A fully penetrating dewatering well in an unconfined sand-and-gravel aquifer (K = 45 ft/day) resting on horizontal impermeable clay is pumped to steady state. Observation wells at r1 = 25 ft and r2 = 150 ft show saturated thicknesses (water level above the clay) of h1 = 38.0 ft and h2 = 41.5 ft. The well radius is 0.5 ft, the static saturated thickness was 44 ft, and the pump is set 5 ft above the clay.

Most nearly, the pumping rate is:

A. 50,551 ft³/day
B. 21,954 ft³/day
C. 967 ft³/day
D. 43,908 ft³/day

Given equations / data

> Unconfined (Dupuit–Thiem): $Q = \pi K(h2^2 - h1^2)/\ln(r2/r1)$

Worked solution

1. $h2^2 - h1^2 = 1722.25 - 1444.00 = 278.25\ ft^2$
2. $\ln(150/25) = 1.7918$
3. $Q = \pi \times 45 \times 278.25/1.7918 = 21{,}954\ ft^3/day\ (\approx 114\ gpm)$

Final answer

21,954 ft³/day

Why the other values are wrong

- **50,551 ft³/day** — used log10
- **967 ft³/day** — squared the difference instead of differencing squares
- **43,908 ft³/day** — used 2π (confined form)

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §6.6.4 Well Analysis–Steady State (unconfined aquifer); §3.16.5 Construction Dewatering

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gpeyt1-29 · Confined aquifer K

GEO PE · Long Calculation Problems I — Topic 6: Groundwater & Seepage

Problem

A pumping test in a 30 ft thick confined sand aquifer (overlain by 22 ft of clay) runs at a steady 350 gpm. Drawdowns: 9.2 ft at r = 40 ft and 5.1 ft at r = 200 ft. The pumping well radius is 0.67 ft; the piezometric surface is 8 ft below grade before pumping.

Most nearly, the hydraulic conductivity K is:

A. 140 ft/day
B. 61 ft/day
C. 40 ft/day
D. 281 ft/day

Given equations / data

> Confined: Q = 2πKb(h2 − h1)/ln(r2/r1); h2 − h1 = s1 − s2; 1 ft³ = 7.48 gal

Worked solution

1. Q = 350 gpm / 7.48 × 1,440 = 67,380 ft³/day
2. h2 − h1 = 9.2 − 5.1 = 4.1 ft
3. K = 67,380 × ln(200/40) / (2π × 30 × 4.1) = 140.3 ft/day

Final answer

140 ft/day

Why the other values are wrong

- **61 ft/day** — used log10
- **40 ft/day** — added drawdowns
- **281 ft/day** — used π instead of 2π

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §6.6.4 Well Analysis–Steady State (confined aquifer); UFC 3-220-05 Ch. 4

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=RqaahrVtV4k (no creator values copied).

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gpeyt1-30 · Layered k (vertical)

GEO PE · Long Calculation Problems I — Topic 6: Groundwater & Seepage

Problem

A 12 ft soil column beneath a detention pond consists of 4 ft of sand (k = 2×10^{-3} cm/s), 2 ft of silty clay (k = 5×10^{-6} cm/s), and 6 ft of silty sand (k = 8×10^{-4} cm/s), top to bottom. The total head loss across the column is 9 ft. Flow is vertical (downward).

Most nearly, the equivalent vertical hydraulic conductivity is:

A. 1.1e-03 cm/s
B. 2.9e-05 cm/s
C. 2.0e-04 cm/s
D. 5.0e-06 cm/s

Given equations / data

> $k_{v,eq} = \Sigma H_i / \Sigma(H_i/k_i)$; $k_{h,eq} = \Sigma(k_i H_i)/\Sigma H_i$

Worked solution

1. $\Sigma\ H_i/k_i = 4/2e\text{-}3 + 2/5e\text{-}6 + 6/8e\text{-}4 = 409{,}500$ ft·s/cm
2. $k_v = 12/409{,}500 = 2.93e\text{-}05$ cm/s
3. ($k_h = 1.07e\text{-}03$ cm/s for comparison; clay layer controls vertical flow)

Final answer

2.9e-05 cm/s

Why the other values are wrong

- **1.1e-03 cm/s** — used the horizontal (arithmetic) average
- **2.0e-04 cm/s** — used a geometric mean
- **5.0e-06 cm/s** — took the clay k alone

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.16.3 Flow Through Soil (layered/equivalent k), p. 188

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=RqaahrVtV4k (no creator values copied).

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gpeyt1-31 · Corrosion allowance

GEO PE · Long Calculation Problems I — Topic 7: Problematic Soil & Rock Conditions

Problem

Uncoated carbon-steel H-piles will be driven through a fill with field resistivity 1,500 Ω·cm (pH 5.8, sulfates 400 ppm) for a bridge with a 75-year design life. The owner requires the maximum (conservative) loss rate per face. Flange thickness is 0.61 in.

Most nearly, the sacrificial thickness loss per exposed face over the design life is:

A. 0.099 in.
B. 0.049 in.
C. 0.016 in.
D. 0.187 in.

Given equations / data

> x = K tⁿ (μm); carbon steel n = 0.80, K = 40 (avg) / 80 (max); 1,000 μm = 1 mm = 0.039 in.

Worked solution

1. 1,500 Ω·cm → corrosive (700–2,000) per resistivity table
2. x = 80 × 75^0.80 = 2,530 μm
3. = 2.53 mm × 0.039 = 0.099 in. per face

Final answer

0.099 in.

Why the other values are wrong

- **0.049 in.** — used the average K = 40
- **0.016 in.** — used galvanized-steel constants
- **0.187 in.** — multiplied t by n instead of raising to n

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.17.1 Reactive and Corrosive Soils (resistivity table), p. 197; §3.17.2 Corrosion of Buried Steel, p. 198

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=ZOpOWZXWblU (no creator values copied).

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gpeyt1-32 · Rock slope with joint water

GEO PE · Long Calculation Problems I — Topic 7: Problematic Soil & Rock Conditions

Problem

A sandstone cut contains a planar bedding joint dipping α = 38° (length L = 55 ft) and a vertical tension crack behind the crest. After a storm the tension crack fills to hw = 18 ft with water that drains along the joint to the face. Block weight W = 210,000 lb/ft. Joint c = 300 psf, φ = 32°. No seismic load. Rock RMR = 58 (fair).

Most nearly, the factor of safety with the crack full of water is:

A. 0.93
B. 0.85
C. 0.78
D. 0.70

Given equations / data

> FS = [cL + (W cos α − PWR − PWJ sin α) tan φ]/(W sin α + PWJ cos α); PWJ = ½γw hw²; PWR = ½γw hw L

Worked solution

1. PWJ = ½(62.4)(18)² = 10,109 lb/ft
2. PWR = ½(62.4)(18)(55) = 30,888 lb/ft
3. N′ = 165,482 − 30,888 − 6,224 = 128,371
4. Resisting = 16,500 + N′ tan32 = 96,715
5. Driving = 129,289 + 7,966 = 137,255
6. FS = 0.70 (dry 0.93)

Final answer

0.70

Why the other values are wrong

- **0.93** — ignored water forces
- **0.85** — omitted uplift PWR on the joint
- **0.78** — omitted the tension-crack force PWJ

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.6.3 Rock Slope Failure, p. 104

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=8cX88YWUo4U, https://www.youtube.com/watch?v=jvEQeZuqydI (no creator values copied).

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gpeyt1-33 · Galvanized coating life

GEO PE · Long Calculation Problems I — Topic 7: Problematic Soil & Rock Conditions

Problem

MSE wall steel strips are hot-dip galvanized with a 86 μm zinc coating. Backfill resistivity tests average 4,200 Ω·cm, pH 7.1, chlorides 60 ppm. Estimate how long the zinc lasts using the AVERAGE galvanized loss constants before the base steel starts to corrode.

Most nearly, the zinc coating life is:

A. 7 years
B. 2 years
C. 5 years
D. 3 years

Given equations / data

> $x = K\,t^{n}$; galvanized: $n = 0.65$, $K = 25$ (avg) / 50 (max) μm

Worked solution

1. 4,200 Ω·cm → moderately corrosive (2,000–5,000)
2. t = (x/K)^(1/n) = (86/25)^(1/0.65) = 3.44^1.538 = 6.7 yr

Final answer

7 years

Why the other values are wrong

- **2 years** — used the maximum K = 50
- **5 years** — divided by n instead of taking the 1/n root
- **3 years** — used carbon-steel constants

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.17.1–3.17.2, pp. 197–198

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=ZOpOWZXWblU (no creator values copied).

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gpeyt1-34 · Collapsible soil settlement

GEO PE · Long Calculation Problems I — Topic 7: Problematic Soil & Rock Conditions

Problem

A 9 ft thick layer of dry loess will support a spread-footing warehouse. A double-oedometer test on a sample from mid-layer (initial e0 = 0.98) shows: under the design stress (dry) e = 0.94; after inundation at that same stress e = 0.83. Natural w = 6%, γd = 86 pcf. The owner asks how much additional settlement occurs if lawn irrigation wets the layer after construction.

Most nearly, the collapse settlement caused by wetting is:

A. 8.2 in.
B. 11.9 in.
C. 6.0 in.
D. 0.5 in.

Given equations / data

$$\Delta H = H\,\Delta e/(1 + e0)$$

Worked solution

1. Δe (collapse) $= 0.94 - 0.83 = 0.11$
2. $\Delta H = 9 \times 0.11/(1 + 0.98) = 0.500$ ft $= 6.0$ in.

Final answer

6.0 in.

Why the other values are wrong

- **8.2 in.** — included the dry load-compression as collapse
- **11.9 in.** — forgot to divide by (1 + e0)
- **0.5 in.** — left the answer in feet labeled inches

References (Jayce's library)

- Vandevelde, ASCE PE Review — Foundation Engineering (S = HΔe/(1+e0) worked example); NCEES PE Civil Reference Handbook v2.1 §3.2 Consolidation (strain form)

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=ZOpOWZXWblU, https://www.youtube.com/watch?v=M9uBSVNmUgU (no creator values copied).

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gpeyt1-35 · Secondary compression (organic)

GEO PE · Long Calculation Problems I — Topic 7: Problematic Soil & Rock Conditions

Problem

A 14 ft layer of organic silt (OH, organic content 14%) under a preloaded embankment reaches the end of primary consolidation at t = 1.8 years. The strain-based coefficient of secondary compression is Cα = 0.012. The design life of the roadway is 40 years from construction. Primary settlement was 22 in.

Most nearly, the secondary settlement from end of primary to end of design life is:

A. 3.2 in.
B. 6.3 in.
C. 1.0 in.
D. 2.7 in.

Given equations / data

> Ss = Cα H0 Δlog t (Cα in strain form)

Worked solution

1. Δlog t = log(40/1.8) = 1.3468
2. Ss = 0.012 × 14 × 1.3468 = 0.2263 ft = 2.7 in.

Final answer

2.7 in.

Why the other values are wrong

- **3.2 in.** — used log t2 (started the clock at 1 yr)
- **6.3 in.** — used natural log
- **1.0 in.** — divided by (1 + e0) again although Cα is in strain form

References (Jayce's library)

- Vandevelde, ASCE PE Review — Foundation Engineering (Ss = Cα H0 Δlog t)

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=IGS94AWmPWA (no creator values copied).

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gpeyt1-36 · Rankine thrust with water table

GEO PE · Long Calculation Problems I — Topic 8: Retaining Structures

Problem

A 22 ft high vertical-backed wall retains level granular backfill (c = 0, φ = 32°). Because the weep holes are clogged, a water table has built up 8 ft below the top of the backfill. γ = 118 pcf above the water table, γsat = 126 pcf below. Ignore wall friction; the wall moves enough to mobilize active pressure. The concrete stem is 18 in. thick.

Most nearly, the total horizontal force (soil + water) per foot of wall is:

A. 7,136 lb/ft
B. 13,251 lb/ft
C. 9,015 lb/ft
D. 8,774 lb/ft

Given equations / data

> $K_a = \tan^2(45 - \varphi/2)$; $\sigma'_h = K_a \sigma'_v$; $u = \gamma_w h_w$; P = area of pressure diagram

Worked solution

1. $K_a = 0.3073$
2. σ'_h at 8 ft = $0.3073 \times 118 \times 8 = 290$ psf
3. σ'_h at 22 ft = $290 + 0.3073 \times 63.6 \times 14 = 564$ psf
4. Soil: $\tfrac{1}{2}(290)(8) + 290(14) + \tfrac{1}{2}(274)(14) = 7{,}136$ lb/ft
5. Water: $\tfrac{1}{2}(62.4)(14)^2 = 6{,}115$ lb/ft
6. Total = 13,251 lb/ft

Final answer

13,251 lb/ft

Why the other values are wrong

- **7,136 lb/ft** — forgot the water pressure
- **9,015 lb/ft** — used γsat (total) below the WT and no separate water
- **8,774 lb/ft** — treated the backfill as dry

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.1.2 Rankine Earth Coefficients, p. 72

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=w48hBSXrUhY (no creator values copied).

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gpeyt1-37 · Overturning FS

GEO PE · Long Calculation Problems I — Topic 8: Retaining Structures

Problem

A 12 ft tall plain-concrete gravity wall (γc = 150 pcf) has a vertical back face and a battered front face. Crest width 2 ft, base width 7 ft (toe at the front). Backfill is level, dry sand: γ = 120 pcf, φ = 30°, c = 0, no wall friction. Passive resistance in front is neglected. The foundation is dense gravel with δ = 28°.

Most nearly, the factor of safety against overturning is:

A. 2.12
B. 0.96
C. 3.18
D. 1.59

Given equations / data

> Ka = tan²(45 − φ/2); Pa = ½KaγH² at H/3; FSOT = ΣMresisting/ΣMoverturning about the toe

Worked solution

1. Ka = 1/3; Pa = ½(1/3)(120)(12)² = 2,880 lb/ft at 4.0 ft
2. Rectangle (back): W1 = 2×12×150 = 3,600 lb at 6.00 ft from toe
3. Triangle (front): W2 = ½×5×12×150 = 4,500 lb at 3.33 ft
4. MR = 36,600; MO = 2,880×4.0 = 11,520 ft-lb/ft
5. FSOT = 3.18

Final answer

3.18

Why the other values are wrong

- **2.12** — applied Pa at H/2
- **0.96** — took moments about the heel-side distances (wrong lever arms)
- **1.59** — forgot the ½ in Pa

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.1.2 Rankine; §3.18 / wall stability figures (moments about toe), p. 209

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=P-W_f8IyVos (no creator values copied).

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gpeyt1-38 · Sliding FS with passive

GEO PE · Long Calculation Problems I — Topic 8: Retaining Structures

Problem

A cantilever wall has total vertical load (concrete + soil on heel) ΣV = 15,400 lb/ft and horizontal active thrust Ph = 6,200 lb/ft. The base is 9 ft wide and cast on sand with base friction δb = 26° (ca = 0). Soil in front of the toe is 3 ft deep (γ = 120 pcf, φ = 30°) and the designer will count full Rankine passive resistance on the toe. Overturning FS was already checked at 2.4.

Most nearly, the factor of safety against sliding is:

A. 1.21
B. 1.47
C. 1.24
D. 1.73

Given equations / data

> FSs = [(ΣV) tan δb + ca B + Pp]/Ph; Kp = tan²(45 + φ/2); Pp = ½Kpγ Df²

Worked solution

1. Base friction = 15,400 × tan26 = 7,511 lb/ft
2. Kp = 3.00; Pp = ½(3.00)(120)(3)² = 1,620 lb/ft
3. FSs = (7,511 + 1,620)/6,200 = 1.47

Final answer

1.47

Why the other values are wrong

- **1.21** — ignored passive resistance
- **1.24** — used Ka instead of Kp in front
- **1.73** — forgot the ½ in Pp

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 Factor of Safety against Sliding (cantilever/counterfort wall figure), p. 209; §3.1.2 Rankine

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=90zXmI_yn_g, https://www.youtube.com/watch?v=F2golrHWDO4 (no creator values copied).

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gpeyt1-39 · Uniform surcharge thrust

GEO PE · Long Calculation Problems I — Topic 8: Retaining Structures

Problem

A 16 ft basement-type wall that can yield to the active state retains dry, level silty sand (γ = 115 pcf, φ = 30°, c = 0). A uniform surcharge of q = 400 psf from a parking area acts over the full backfill. No water; ignore wall friction. The designer needs the total thrust and its height above the base for the moment check.

Most nearly, the total thrust and its height above the base are:

A. 7,040 lb/ft at 6.1 ft
B. 7,040 lb/ft at 5.3 ft
C. 11,307 lb/ft at 6.8 ft
D. 5,973 lb/ft at 5.3 ft

Given equations / data

> $\sigma h = K_a(q + \gamma z)$; $P_q = K_a q H$ at $H/2$; $P_s = \frac{1}{2} K_a \gamma H^2$ at $H/3$

Worked solution

1. $K_a = 1/3$
2. $P_q = (1/3)(400)(16) = 2{,}133$ lb/ft at 8 ft
3. $P_s = \frac{1}{2}(1/3)(115)(16)^2 = 4{,}907$ lb/ft at 5.33 ft
4. $P = 7{,}040$ lb/ft; $\bar{y} = (2{,}133 \times 8 + 4{,}907 \times 5.33)/7{,}040 = 6.14$ ft

Final answer

7,040 lb/ft at 6.1 ft

Why the other values are wrong

- **7,040 lb/ft at 5.3 ft** — placed the whole resultant at H/3
- **11,307 lb/ft at 6.8 ft** — applied full q (forgot Ka on the surcharge)
- **5,973 lb/ft at 5.3 ft** — treated the surcharge as triangular

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.1.2 Rankine; §3.13.5.1 $\sigma_{hq} = K_a q$; §3.1.4 Load Distribution from Surcharge

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=4cCtjNpcFkA (no creator values copied).

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gpeyt1-40 · Wall base pressure

GEO PE · Long Calculation Problems I — Topic 8: Retaining Structures

Problem

For a cantilever retaining wall on medium dense sand, the resultant of all vertical forces ΣV = 18,200 lb/ft (Pv included) intersects the 10 ft base at d = 4.1 ft from the toe, computed from moments about the toe. The allowable bearing pressure for the sand is 4,000 psf. Overturning FS = 2.3, sliding FS = 1.6.

Most nearly, the maximum toe pressure qmax is:

A. 2,220 psf
B. 2,803 psf
C. 837 psf
D. 1,984 psf

Given equations / data

> `e = B/2 − d; qmax = (ΣV/B)(1 + 6e/B); qeq (Meyerhof) = ΣV/B′, B′ = B − 2e; for soils e ≤ B/6`

Worked solution

1. `e = 10/2 − 4.1 = 0.90 ft ≤ B/6 = 1.67 ✓`
2. `qmax = (18,200/10)(1 + 6×0.90/10) = 2,803 psf`
3. `qmin = 837 psf; qeq = 18,200/8.2 = 2,220 psf (for comparison)`

Final answer

2,803 psf

Why the other values are wrong

- **2,220 psf** — reported Meyerhof uniform qeq
- **837 psf** — reported qmin (heel)
- **1,984 psf** — used (1 + e/B), dropped the 6

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 Applied Stress at Base (qmax, qmin, qeq) and eccentricity criteria (cantilever wall figure), p. 209

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=P-W_f8IyVos (no creator values copied).

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gpeyt1-41 · Square footing, WT at base

GEO PE · Long Calculation Problems I — Topic 9: Shallow Foundations

Problem

A 6 ft × 6 ft column footing bears 4 ft below grade in a uniform medium-dense sand (c = 0, φ = 32°, γ = 120 pcf above and γsat ≈ 125 pcf below the water table). The seasonal-high water table is at the footing base (Dw = Df). Use the AASHTO/NHI general bearing equation with handbook shape factors and groundwater correction factors, FS = 3.0. Column dead load 160 kips, live load 90 kips (not needed to compute capacity).

Most nearly, the allowable column load (gross, FS = 3) is:

A. 295 kips
B. 256 kips
C. 199 kips
D. 148 kips

Given equations / data

> qult = cNc sc + qNq sq Cwq + 0.5γBNγ sγ Cwγ; φ = 32°: Nq = 23.2, Nγ = 30.2; sq = 1 + (B/L) tan φ; sγ = 1 − 0.4(B/L); Dw = Df → Cwq = 1.0, Cwγ = 0.5

Worked solution

1. q = 120×4 = 480 psf; sq = 1 + tan32 = 1.625; sγ = 0.6
2. Surcharge term = 480×23.2×1.625×1.0 = 18,095 psf
3. γ term = 0.5×120×6×30.2×0.6×0.5 = 3,262 psf
4. qult = 21,356 psf; qall = 7,119 psf
5. Qall = 7,119 × 36 ft² = 256 kips

Final answer

256 kips

Why the other values are wrong

- **295 kips** — ignored the groundwater correction (Cwγ = 1)
- **199 kips** — omitted shape factors
- **148 kips** — also took Cwq = 0.5 (misread the Dw = Df row)

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.4.2–3.4.2.1 Bearing Capacity Equation, Shape Factors, Bearing Capacity Factors and Groundwater Correction tables, pp. 90–91

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=du7B1MO-GCg, https://www.youtube.com/watch?v=ZUGdBao3tkc, https://www.youtube.com/watch?v=1fx03dLbmQc (no creator values copied).

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gpeyt1-42 · Strip footing on clay (φ = 0)

GEO PE · Long Calculation Problems I — Topic 9: Shallow Foundations

Problem

A 5 ft wide continuous wall footing bears 3.5 ft deep in a saturated, homogeneous stiff clay. Unconfined compression tests give an average qu = 2,400 psf; γ = 118 pcf; the water table is deep. Analyze the short-term (undrained) case with FS = 3.0 on net bearing pressure. The wall load is 9 kips/ft.

Most nearly, the allowable NET bearing pressure is:

A. 4,112 psf
B. 2,194 psf
C. 2,280 psf
D. 2,056 psf

Given equations / data

> φ = 0: Nc = 5.14, Nq = 1, Nγ = 0; su = qu/2; qnet,ult = qult − γDf = 5.14 su

Worked solution

1. su = 2,400/2 = 1,200 psf
2. qnet,ult = 5.14 × 1,200 = 6,168 psf
3. qnet,all = 6,168/3.0 = 2,056 psf
4. (check: wall pressure 9,000/5 = 1,800 psf gross)

Final answer

2,056 psf

Why the other values are wrong

- **4,112 psf** — used qu as su (forgot ÷2)
- **2,194 psf** — reported gross (added γDf after FS)
- **2,280 psf** — used Terzaghi Nc = 5.7

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.4.2 Bearing Capacity Equation (strip) and factor table (φ = 0: Nc = 5.14), pp. 90–91; §3.3.3 Undrained Shear Strength

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=l9jr9yfBzG0, https://www.youtube.com/watch?v=ZUGdBao3tkc (no creator values copied).

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gpeyt1-43 · Eccentric footing (Meyerhof)

GEO PE · Long Calculation Problems I — Topic 9: Shallow Foundations

Problem

An 8 ft × 8 ft footing supports a vertical load P = 180 kips (including footing weight) and an overturning moment M = 120 kip-ft about one axis from wind on a sign structure. The footing bears on dense sand where settlement analysis uses the Meyerhof equivalent uniform pressure. Allowable bearing pressure is 4.0 ksf.

Most nearly, the Meyerhof equivalent uniform bearing pressure is:

A. 4,219 psf
B. 2,812 psf
C. 3,375 psf
D. 3,068 psf

Given equations / data

> $e = M/P \ (\le B/6)$; $B' = B - 2e$; $q_{eq} = P/(B'L)$; $q_{max} = (P/BL)(1 + 6e/B)$

Worked solution

1. $e = 120/180 = 0.667$ ft $\le B/6 = 1.33$ ✓
2. $B' = 8 - 2(0.667) = 6.667$ ft
3. $q_{eq} = 180/(6.667 \times 8) = 3.375$ ksf $= 3{,}375$ psf
4. ($q_{max} = 4{,}219$ psf)

Final answer

3,375 psf

Why the other values are wrong

- **4,219 psf** — reported qmax (trapezoidal toe pressure)
- **2,812 psf** — ignored eccentricity
- **3,068 psf** — used $B' = B - e$

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 Applied stress at base — $q_{eq} = \Sigma V/B'$, $B' = B - 2e$ (cantilever wall figure, p. 209); NHI-06-089 §8 effective width

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=du7B1MO-GCg (no creator values copied).

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gpeyt1-44 · 2:1 stress + clay settlement

GEO PE · Long Calculation Problems I — Topic 9: Shallow Foundations

Problem

A 10 ft × 10 ft footing with net applied pressure Δp = 2,600 psf bears 3 ft below grade on sand (γ = 120 pcf above and below the water table, which is at the footing base). A 10 ft normally consolidated clay layer (γsat = 112 pcf, e0 = 0.92, Cc = 0.3) begins 10 ft below the footing base. Use the 2:1 method at mid-clay and a single layer.

Most nearly, the consolidation settlement of the clay is:

A. 9.5 in.
B. 3.6 in.
C. 2.5 in.
D. 1.5 in.

Given equations / data

> $\Delta\sigma_z = \Delta p\, B\, L/[(B + z)(L + z)]$ (2:1, z below footing base); $S = C_c H/(1 + e_0) \log(\sigma'_f/\sigma'_0)$; $\gamma_w = 62.4$ pcf

Worked solution

1. z (base → mid-clay) = 10 + 5 = 15 ft
2. $\Delta\sigma = 2{,}600 \times 10^2/(10+15)^2 = 416$ psf
3. $\sigma'_0 = 3(120) + 10(120-62.4) + 5(112-62.4) = 1{,}184$ psf
4. $S = 0.3 \times 10/1.92 \times \log(1{,}600/1{,}184) = 0.2043$ ft = 2.5 in.

Final answer

2.5 in.

Why the other values are wrong

- **9.5 in.** — applied full Δp at mid-clay (no spreading)
- **3.6 in.** — evaluated Δσ at the top of clay
- **1.5 in.** — used total stress for σ′0

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.5.1 Stress Distribution (2:1 method), p. 94; §3.2.1 NC consolidation, p. 83

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=NwGztxby-vQ, https://www.youtube.com/watch?v=SLtD_6vcRAs (no creator values copied).

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gpeyt1-45 · Elastic settlement

GEO PE · Long Calculation Problems I — Topic 9: Shallow Foundations

Problem

A flexible 9 ft × 9 ft mat for an equipment pad applies Δp = 3,200 psf to a deep, homogeneous dense gravelly sand with Em = 420,000 psf (≈ 2,917 psi) and ν = 0.3. Estimate the AVERAGE immediate settlement under the flexible pad (not the center).

Most nearly, the average elastic settlement is:

A. 0.71 in.
B. 0.84 in.
C. 0.78 in.
D. 0.06 in.

Given equations / data

> δv = Cd Δp Bf (1 − ν²)/Em; Cd (square, flexible): center 1.12, corner 0.56, average 0.95; square rigid 0.99

Worked solution

1. 1 − ν² = 0.91
2. δv = 0.95 × 3,200 × 9 × 0.91/420,000 = 0.0593 ft = 0.71 in.

Final answer

0.71 in.

Why the other values are wrong

- **0.84 in.** — used the center factor 1.12
- **0.78 in.** — omitted (1 − ν²)
- **0.06 in.** — left the answer in ft labeled in.

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.5.2 Settlement (Elastic Method) and Cd table, p. 95

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=SLtD_6vcRAs (no creator values copied).

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gpeyt1-46 · Driven pile in clay (α)

GEO PE · Long Calculation Problems I — Topic 10: Deep Foundations

Problem

A 16 in. square precast concrete pile is driven 50 ft into a two-layer clay deposit: 0–20 ft soft clay su = 600 psf (α = 0.95); 20–60 ft stiff clay su = 1400 psf (α = 0.62). The water table is at 5 ft. Clay γsat = 115 pcf. Use the α-method for side resistance and qp = 9su at the toe. Design load 90 kips; ASD FS = 2.5.

Most nearly, the ultimate axial compression capacity Ru is:

A. 310 kips
B. 200 kips
C. 174 kips
D. 222 kips

Given equations / data

> Rs = Σ α su (perimeter)(ΔL); Rt = 9 su At; Ru = Rs + Rt

Worked solution

1. Perimeter = 4×1.333 = 5.333 ft; At = 1.778 ft²
2. Layer 1: 0.95×600×5.333×20 = 60,800 lb
3. Layer 2: 0.62×1400×5.333×30 = 138,880 lb
4. Rt = 9×1400×1.778 = 22,400 lb
5. Ru = 222,080 lb ≈ 222 kips (Rall = 89 kips at FS 2.5 > 90 ✓)

Final answer

222 kips

Why the other values are wrong

- **310 kips** — omitted α (α = 1)
- **200 kips** — left out toe resistance
- **174 kips** — treated the pile as a 16 in. round pile

References (Jayce's library)

- FHWA-NHI-16-009 (GEC 12) Vol I §7.2 α-method (Eq. 7-31) and qp = 9su (Eq. 7-16)

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=hfbcwIJfAjw, https://www.youtube.com/watch?v=Jdcs8-i_OA4, https://www.youtube.com/watch?v=ZcjQbEB66IQ (no creator values copied).

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gpeyt1-47 · Driven pile in sand (β)

GEO PE · Long Calculation Problems I — Topic 10: Deep Foundations

Problem

A 14 in. diameter closed-end steel pipe pile is driven 45 ft into medium-dense sand. Water table at 10 ft; γ = 115 pcf above, γsat = 125 pcf below. Use the effective-stress β-method with β = 0.38 along the full length and toe coefficient Nt = 40 (from the Fellenius ranges for this sand). Ignore any limiting values. SPT N60 ≈ 22; φ′ ≈ 34°.

Most nearly, the ultimate capacity Ru is:

A. 407 kips
B. 260 kips
C. 117 kips
D. 352 kips

Given equations / data

> Rs = β ∫σ′v dA (use average σ′v of each segment × side area); Rt = Nt σ′v,toe At; γw = 62.4 pcf

Worked solution

1. σ′v at 10 ft = 1,150 psf; at toe = 1,150 + 35(125−62.4) = 3,341 psf
2. Perimeter = π(1.167) = 3.665 ft; At = 1.069 ft²
3. Σσ′v·L = ½(1,150)(10) + ½(1,150+3,341)(35) = 84,342 lb/ft
4. Rs = 0.38×3.665×that = 117,470 lb
5. Rt = 40×3,341×1.069 = 142,863 lb
6. Ru = 260 kips

Final answer

260 kips

Why the other values are wrong

- **407 kips** — used total stress (no pore-pressure subtraction)
- **117 kips** — left out toe resistance
- **352 kips** — used toe σ′v along the whole shaft

References (Jayce's library)

- FHWA-NHI-16-009 (GEC 12) Vol I §7.2 effective-stress (β) method, Rs and Rt with Nt

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=Jdcs8-i_OA4, https://www.youtube.com/watch?v=hfbcwIJfAjw (no creator values copied).

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gpeyt1-48 · Uplift capacity in clay

GEO PE · Long Calculation Problems I — Topic 10: Deep Foundations

Problem

A transmission tower leg is founded on an 18 in. diameter drilled concrete pier, 40 ft long, in uniform stiff clay (su = 1,100 psf, adhesion factor α = 0.7). The water table is deep (below the pier). Concrete γc = 150 pcf. Apply FS = 3.0 to side resistance only; pier self-weight is counted at full value (unfactored). Ignore any belling and end suction.

Most nearly, the allowable uplift capacity is:

A. 52 kips
B. 59 kips
C. 48 kips
D. 55 kips

Given equations / data

```
Tall = α su π D L / FS + Wpier
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Worked solution

1. Qs = 0.7×1,100×π×1.50×40 = 145,142 lb
2. W = π/4×1.50²×40×150 = 10,603 lb
3. Tall = 145,142/3.0 + 10,603 = 58,983 lb ≈ 59 kips

Final answer

59 kips

Why the other values are wrong

- **52 kips** — applied FS to pier weight too
- **48 kips** — ignored pier weight
- **55 kips** — used buoyant concrete weight though WT is deep

References (Jayce's library)

- FHWA-NHI-16-009 (GEC 12) α-method side resistance; uplift Rn = Rs + W (Eq. 7-41 group form)

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=jnAwn45noZw (no creator values copied).

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gpeyt1-49 · Pile group block failure

GEO PE · Long Calculation Problems I — Topic 10: Deep Foundations

Problem

A 3 × 3 group of 12 in. square piles at 3 ft center-to-center spacing is driven 45 ft into soft-to-medium clay with weighted average su1 = 900 psf along the embedment and su2 = 1,600 psf below the toe (to 2B). Each single pile has nominal capacity 95 kips (α-method + toe). Pile cap is not in contact with the ground.

Most nearly, the nominal group capacity is:

A. 1,840 kips
B. 855 kips
C. 1,490 kips
D. 1,273 kips

Given equations / data

> Block: Rng = 2D(B + Z)su1 + B Z su2 Nc; Nc = 5(1 + D/5B)(1 + B/5Z) ≤ 9; B = Z = outside-to-outside group width; group capacity = lesser of block and Σ singles

Worked solution

1. B = Z = 2(3) + 1 = 7 ft; Nc = 5(1 + 45/35.0)(1.2) = 13.7 → cap at 9
2. Rblock = 2(45)(7.0+7.0)(900) + 7.0²(1,600)(9) = 1,839,600 lb
3. Σ singles = 9 × 95 = 855 kips
4. Group = lesser = 855 kips

Final answer

855 kips

Why the other values are wrong

- **1,840 kips** — took the larger value
- **1,490 kips** — used center-to-center (6 ft) group width for block
- **1,273 kips** — forgot the 2 on perimeter shear

References (Jayce's library)

- FHWA-NHI-16-009 (GEC 12) Vol I Eq. 7-36 and 7-37 (block failure, Nc)

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=MI4gaUwboyE (no creator values copied).

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gpeyt1-50 · Pile group settlement (equivalent footing)

GEO PE · Long Calculation Problems I — Topic 10: Deep Foundations

Problem

A pile group with plan dimensions 8 ft × 8 ft (outside-to-outside) carries Q = 900 kips. The friction piles are 45 ft long in a stiff clay crust that overlies a 20 ft normally consolidated clay layer from 45 ft to 65 ft (γsat = 118 pcf, e0 = 1.0, Cc = 0.36). Water table at the ground surface; stiff crust also γsat = 118 pcf. Place the equivalent footing at 2/3 of the embedded length and spread load at 2:1; evaluate one layer at mid-depth of the NC clay.

Most nearly, the consolidation settlement of the group is:

A. 4.5 in.
B. 12.1 in.
C. 1.3 in.
D. 2.2 in.

Given equations / data

> Equivalent footing at D′ = (2/3)D; Δσ = Q/[(B + z)(L + z)] with z measured from D′; S = Cc H/(1 + e0) log(σ′f/σ′0)

Worked solution

1. D′ = (2/3)(45) = 30 ft; mid-clay depth = 55 ft → z = 25 ft
2. Δσ = 900,000/(8+25)² = 826 psf
3. σ′0 = 55(118−62.4) = 3,058 psf
4. S = 0.36×20/2.0 × log(3,884/3,058) = 4.5 in.

Final answer

4.5 in.

Why the other values are wrong

- **12.1 in.** — placed the equivalent footing at the pile toe
- **1.3 in.** — spread the load from the ground surface
- **2.2 in.** — used total stress for σ′0

References (Jayce's library)

- FHWA-NHI-16-009 (GEC 12) Vol I §7 equivalent-footing method (2/3 D)
- NCEES PE Civil Reference Handbook v2.1 §3.5.1 2:1 method; §3.2.1

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=MI4gaUwboyE (no creator values copied).

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